

Mathematical Methods I

The metric and tensors

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1. Tensors and their components

A tensor generalizes the notions of a scalar and a vector. In a coordinate system, its components are identified by indices; the total number of indices is its **rank**.

Types of tensor

- **Rank 0:** a scalar, such as mass or temperature at a point.
- **Rank 1:** a vector, such as velocity or force, or a covector, such as the differential of a function.
- **Rank 2 or higher:** a tensor whose components form a matrix or a higher dimensional array.

In dimension d , a rank n tensor has d^n components in a basis, although symmetries may reduce the number of independent components.

The defining feature of a tensor is its **transformation law**: its components change in a prescribed way when coordinates change, while the geometric object they represent remains the same.

Geometric definition

At a point p , a tensor of type (r, s) is a multilinear map

$$T_p : \underbrace{T_p^*M \times \cdots \times T_p^*M}_{r \text{ covectors}} \times \underbrace{T_pM \times \cdots \times T_pM}_{s \text{ vectors}} \longrightarrow \mathbb{R}.$$

Its components carry r upper indices and s lower indices. For example, A^i is contravariant, A_i is covariant, and T^i_j is mixed.

2. The metric: from vectors to covectors

Tangent vectors and covectors belong to different spaces. A **metric** g assigns a real number to each pair of tangent vectors, bilinearly and symmetrically. Because it is nondegenerate, it associates each vector v with the covector $v^b = g(v, \cdot)$; this operation is called **lowering an index**.

Covariant metric tensor

In coordinates $x^1, \dots, x^d$, let $e_i = \partial/\partial x^i$ be the tangent basis. The metric components are

$$g_{ij} = g(e_i, e_j) = \left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle, \quad g_{ij} = g_{ji}.$$

The matrix (g_{ij}) represents a symmetric tensor of type $(0, 2)$. In the Euclidean or Riemannian case, $g(v, v) > 0$ whenever $v \neq 0$.

The components $v_i = g_{ij}v^j$ are those of v^b ; here and below, repeated indices are summed. The **inverse metric** has components g^{ij} and satisfies

$$g^{ik}g_{kj} = \delta^i_j, \quad v^i = g^{ij}v_j.$$

Thus g^{ij} is symmetric and of type $(2, 0)$, and it raises an index. The Kronecker delta δ^i_j is the identity matrix.

3. What ds^2 represents

The symbol ds^2 denotes the **line element**: the metric evaluated on an infinitesimal displacement. In Riemannian geometry, it is the square of the infinitesimal distance between nearby points. It is also called the **first fundamental form**.

First fundamental form

$$ds^2 = g_{ij} dx^i dx^j.$$

The Einstein convention is used: repeated indices i and j are summed. The products $dx^i dx^j$ express the quadratic evaluation of the metric on a displacement; they are not exterior products $dx^i \wedge dx^j$.

The metric in action

For tangent vectors $v = v^i e_i$ and $w = w^i e_i$, the inner product and Riemannian norm are

$$\langle v, w \rangle = g_{ij} v^i w^j, \quad \|v\| = \sqrt{g_{ij} v^i v^j}.$$

For nonzero vectors, the angle ϕ is determined by

$$\cos \phi = \frac{g_{ij} v^i w^j}{\sqrt{g_{ij} v^i v^j} \sqrt{g_{ij} w^i w^j}}.$$

If $x^i(t)$ parametrizes a curve with a Riemannian metric, its length from $t = a$ to $t = b$ is obtained

by integrating the speed measured by g :

$$L = \int_a^b \sqrt{g_{ij}(x(t)) \frac{dx^i}{dt} \frac{dx^j}{dt}} dt.$$

In dimension d , the metric also determines the volume element $dV_g = \sqrt{\det(g_{ij})} dx^1 \cdots dx^d$ in oriented Riemannian coordinates. It therefore enters the measurement of both curves and regions.

4. From the tensor product to the wedge product

Tensor product and wedge product

For covectors α, β , the tensor product is defined by

$$(\alpha \otimes \beta)(v, w) = \alpha(v)\beta(w).$$

It need not be symmetric or alternating. In contrast,

$$\alpha \wedge \beta = \alpha \otimes \beta - \beta \otimes \alpha.$$

Thus $dx^1 \otimes dx^1$ is nonzero, although $dx^1 \wedge dx^1 = 0$.

The identity shows what changes when the two vectors are exchanged. Evaluating the products on v, w gives

Evaluation and alternation

$$(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v).$$

Thus $(\alpha \wedge \beta)(w, v) = -(\alpha \wedge \beta)(v, w)$ and $\alpha \wedge \alpha = 0$. In contrast, $(dx^1 \otimes dx^1)(v, v) = [dx^1(v)]^2$ can be nonzero.

Example in the plane

If $\alpha = a dx + b dy$ and $\beta = c dx + d dy$, then

$$\alpha \wedge \beta = (ad - bc) dx \wedge dy.$$

In particular, for $u = (u^x, u^y)$ and $v = (v^x, v^y)$,

$$(dx \wedge dy)(u, v) = u^x v^y - u^y v^x = \det \begin{pmatrix} u^x & v^x \\ u^y & v^y \end{pmatrix}.$$

The wedge product records oriented area; $dx \otimes dy$ alone gives $u^x v^y$ and does not necessarily change sign when the vectors are exchanged.

5. Example: the metric in polar coordinates

In the Euclidean plane, let $u^1 = r$, $u^2 = \theta$, and

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r > 0.$$

To construct the metric, differentiate each Cartesian coordinate with respect to r and θ :

$$\begin{aligned} \frac{\partial x}{\partial r} &= \cos \theta, & \frac{\partial y}{\partial r} &= \sin \theta, \\ \frac{\partial x}{\partial \theta} &= -r \sin \theta, & \frac{\partial y}{\partial \theta} &= r \cos \theta. \end{aligned}$$

The associated tangent vectors are therefore

$$e_r = (\cos \theta, \sin \theta), \quad e_\theta = (-r \sin \theta, r \cos \theta).$$

The Cartesian metric on the plane is Euclidean. Hence,

$$g_{ab} = e_a \cdot e_b = \sum_{k=1}^2 \frac{\partial x^k}{\partial u^a} \frac{\partial x^k}{\partial u^b}, \quad (x^1, x^2) = (x, y), \quad (u^1, u^2) = (r, \theta).$$

Compute its three independent components separately:

$$\begin{aligned} g_{rr} &= \left(\frac{\partial x}{\partial r} \right)^2 + \left(\frac{\partial y}{\partial r} \right)^2 = \cos^2 \theta + \sin^2 \theta = 1, \\ g_{\theta\theta} &= \left(\frac{\partial x}{\partial \theta} \right)^2 + \left(\frac{\partial y}{\partial \theta} \right)^2 = r^2 \sin^2 \theta + r^2 \cos^2 \theta = r^2, \\ g_{r\theta} = g_{\theta r} &= \frac{\partial x}{\partial r} \frac{\partial x}{\partial \theta} + \frac{\partial y}{\partial r} \frac{\partial y}{\partial \theta} = -r \sin \theta \cos \theta + r \sin \theta \cos \theta = 0. \end{aligned}$$

Metric matrix and line element in polar coordinates

$$(g_{ab}) = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad ds^2 = dr^2 + r^2 d\theta^2.$$

The angular coordinate is converted to length by the scale factor r : at fixed radius, a change $d\theta$ corresponds to a distance $r d\theta$.

6. Detailed calculation of a circle's circumference

We seek the circumference of a circle of fixed radius $R > 0$ using the polar metric $ds^2 = dr^2 + r^2 d\theta^2$.

Step 1. Parametrize the curve

Set $t = \theta$ and make one complete revolution for $0 \leq t \leq 2\pi$:

$$u^1(t) = r(t) = R, \quad u^2(t) = \theta(t) = t.$$

The tangent vector therefore has components

$$\frac{dr}{dt} = 0, \quad \frac{d\theta}{dt} = 1.$$

The first equation says that the curve has no radial motion; the second says that the angle increases by one unit per unit of t .

Step 2. Evaluate the metric on the tangent

In polar coordinates, $g_{rr} = 1$, $g_{r\theta} = g_{\theta r} = 0$, and $g_{\theta\theta} = r^2$. Expanding the Einstein sum gives

$$\begin{aligned} g_{ab} \frac{du^a}{dt} \frac{du^b}{dt} &= g_{rr} \left(\frac{dr}{dt} \right)^2 + 2g_{r\theta} \frac{dr}{dt} \frac{d\theta}{dt} + g_{\theta\theta} \left(\frac{d\theta}{dt} \right)^2 \\ &= (1)(0)^2 + 2(0)(0)(1) + (R^2)(1)^2 = R^2. \end{aligned}$$

The speed measured by the metric is $\sqrt{R^2} = R$ because the radius is positive.

Step 3. Integrate the length

Substitute into the length formula, with limits 0 and 2π :

$$L = \int_0^{2\pi} \sqrt{g_{ab} \dot{u}^a \dot{u}^b} dt = \int_0^{2\pi} R dt = R[t]_0^{2\pi} = 2\pi R.$$

This recovers the usual circumference from the line element in curvilinear coordinates.

7. Transformation and invariance

The line element is the same geometric object in any coordinate system. Under a change from x^i to coordinates x'^a , the differentials satisfy

$$dx^i = \frac{\partial x^i}{\partial x'^a} dx'^a.$$

Substituting them into $ds^2 = g_{ij} dx^i dx^j$ gives the covariant transformation law:

Transformation law of the metric

$$g'_{ab} = \frac{\partial x^i}{\partial x'^a} \frac{\partial x^j}{\partial x'^b} g_{ij}.$$

Although the component matrix changes, $g'_{ab} dx'^a dx'^b = g_{ij} dx^i dx^j$.

From Cartesian to polar coordinates

In the plane, the Cartesian metric matrix is the identity. If $J = \partial(x, y)/\partial(r, \theta)$, then

$$J = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad g_{\text{polar}} = J^T J = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}.$$

This matrix multiplication reproduces the calculation of the products $e_a \cdot e_b$.

8. Pythagoras in curvilinear coordinates

In orthogonal coordinates q^i , the metric matrix is diagonal. For a Riemannian metric, define the **scale factor** $h_i = \sqrt{g_{ii}}$, with no sum over i . In three dimensions,

Orthogonal line element

$$ds^2 = (h_1 dq^1)^2 + (h_2 dq^2)^2 + (h_3 dq^3)^2.$$

This is the Pythagorean theorem expressed through infinitesimal coordinate changes. In the polar plane, $h_r = 1$ and $h_\theta = r$.

9. The semi-Riemannian metric

In Riemannian geometry, the metric is **positive definite**: $g(v, v) > 0$ for every nonzero vector. This lets us interpret ds^2 as the square of an infinitesimal length. In **semi-Riemannian** geometry, the metric remains symmetric and nondegenerate but can have both positive and negative signs. Consequently, $g(v, v)$ may be positive, negative, or zero even when $v \neq 0$.

The central physical example is a **Lorentzian** metric, with one time direction and three spatial directions. In flat Minkowski spacetime, using inertial coordinates and the sign convention $(-, +, +, +)$, the metric matrix is

$$(g_{\mu\nu}) = \text{diag}(-c^2, 1, 1, 1) \quad \text{for coordinates } (t, x, y, z).$$

Interval in Minkowski spacetime

With the sign convention $(-, +, +, +)$ and inertial coordinates (t, x, y, z) ,

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2.$$

Here c is the speed of light. The value of the interval is preserved under coordinate changes.